# DATA260 Homework 7
# Agentic Systems, Security, and Evaluation

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**SID4:** 2604

**Model:** qwen2.5:3b

**Port:** 8804

**Git tag:** hw7

This report documents the implementation, experiments, safety checks, and evidence for all six HW7 parts.

## Part 1: JWT Authentication, RBAC, CRUD, and Linting

Implemented JWT bearer authentication with reader, writer, and moderator roles. Readers can view records, writers can create and update records, and moderators can delete records. Ruff and pre-commit enforce code quality before commits.

### part1-jwt-login-success.png

Screenshot not found: part1-jwt-login-success.png

### part1-jwt-login-401.png

Screenshot not found: part1-jwt-login-401.png

### part1-auth-me-no-token-401.png

Screenshot not found: part1-auth-me-no-token-401.png

### part1-reader-list-200.png

Screenshot not found: part1-reader-list-200.png

### part1-writer-delete-403.png

Screenshot not found: part1-writer-delete-403.png

### part1-moderator-delete-200.png

Screenshot not found: part1-moderator-delete-200.png

### part1-precommit-success.png

Screenshot not found: part1-precommit-success.png

## Part 2: Bounded Multi-Step Agent Loop

Built a bounded agent loop that selects allow-listed vulnerability tools, records assistant and tool messages, supports one repair attempt, and stops at a hard step ceiling.

### part2-agent-loop-results.png

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## Part 3: Reflexion and Retry Comparison

Compared a baseline agent with a Reflexion-style agent. The Reflexion agent stores a short evidence-based reflection and retries only within a fixed retry budget.

### part3-reflexion-results.png

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## Part 4: ReAct/MRKL Bike-Share Analysis

Combined three specialized tools: read-only CSV SQL, pricing-policy retrieval, and a safe arithmetic calculator. The experiment produced both membership-wins and pay-per-use-wins scenarios with citations and assumptions.

### part4-mrkl-results.png

Screenshot not found: part4-mrkl-results.png

## Part 5: Prompt-Injection Defenses

Tested recommendation manipulation, unauthorized SQL modification, and data exfiltration attacks. Untrusted text is treated as data, while deterministic tool-level safety checks enforce the actual security boundary.

### part5-prompt-injection-defense.png

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## Part 6: File Tools and Safety Gates

Implemented read, write, edit, glob, search, and controlled bash tools inside a temporary sandbox. The tests include five successful safe operations, five blocked operations, and the glob-to-search-to-read integration flow.

### part6-file-tools-safety.png

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## Verification and Reproducibility

The machine-readable verification record is stored in reports/hw07/verification.json. Raw experiment outputs are stored in reports/hw07/raw/. The implementation is tagged in Git with the tag hw7.